

C.A.V.E: Cave Assessment and Visualization Equipment

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Description

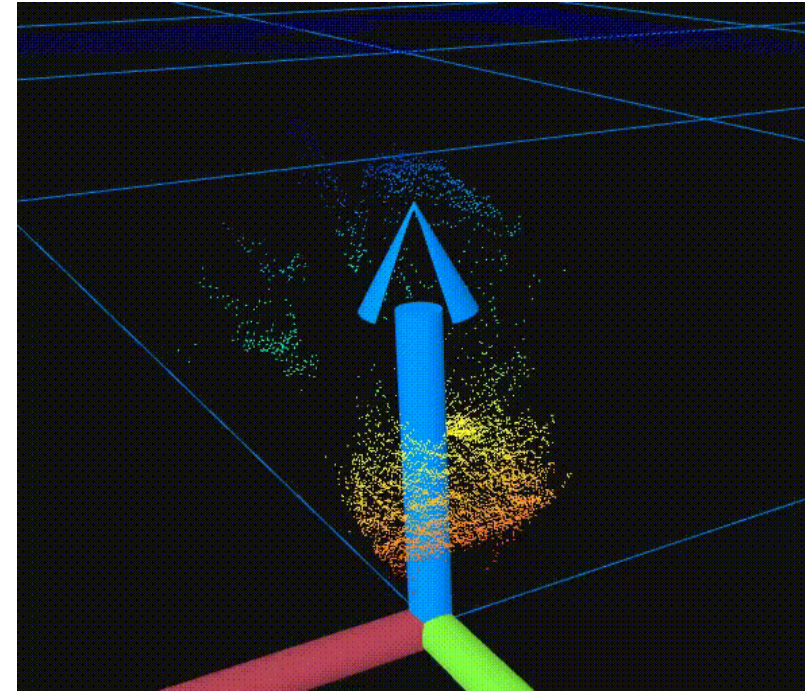
- Map small crevice caves that don't have publicly available layouts
- Difficult to map with current methods, 3-D Lidar is expensive.
- Usage of ToF sensors to get a point cloud of the cave
- Two sensors, mesh their point clouds with ICP
- Get detailed imaging of surrounding area

Proof of Concept

- Record data from the sensor and save into a database file
- Extract the data & apply noise removal filters
- Running the ICP algo on 2 point clouds
- View the results on an external viewer

Further Goals:

- Refine the ICP algorithm if necessary
- Get both sensors working concurrently
- Work with the 2-D Lidar and accelerometer.
- Determine mounting orientations
- Get a detailed point-cloud of a controlled environment



Example Point-Cloud without filters/ICP implemented.